

Chandra Sagar Gurram

AI Group Product Manager · 12+ Years Experience

Career Project Summaries

Problem Statement · Solution · Impact

Interview Preparation Reference · 2025

Table of Contents

1.	Churn Prediction & Retention Engine	Zensar · iGaming
2.	iGaming Fraud Detection Platform	Zensar · iGaming
3.	AI Financial Wellness Chatbot — Old Mutual	Zensar · FinServ
4.	Game Recommendation + Bonus + Demand Forecasting	Zensar · iGaming
5.	FDA-Approved AI Medical Scanner	Photon360
6.	Autonomous Parking — Volley Automation	Photon360
7.	EdTech Fraud Detection	Photon360
8.	Whole Slide Imaging (WSI) Digitization Pipeline	reference · Mayo Clinic
9.	Tumor Segmentation (U-Net + Mask R-CNN)	reference · Mayo Clinic
10.	TIL Assessment — Breast Cancer Immunotherapy	reference · Mayo Clinic
11.	PAIP 2020 — MSI Prediction (Ranked 11th Globally)	Onward Health
12.	Skin Lesion Diagnosis (>93% Accuracy)	Onward Health
13.	PD-L1 Detection — Lung Adenocarcinoma	Onward Health
14.	MSI/MSS Biomarker Classification	Onward Health
15.	OPD Patient Flow Forecasting	AlgoLeap Technologies
16.	CCTV Analytics — GMR Airport	AlgoLeap Technologies
17.	Stock Momentum Prediction Engine	Karvy Stock Broking
18.	Cross-Exchange Arbitrage (LME + CME)	Proprietary Trading

1. Churn Prediction & Retention Engine

Zensar · iGaming Client · \$15M Revenue Impact

The iGaming client was losing high-lifetime-value players with no early warning system — retention was entirely reactive, triggered only after a player had already disengaged, making interventions too late and bonus spend untargeted. I led the design and delivery of a predictive churn engine that identified player disengagement signals 7–14 days before the actual churn event, using an ensemble of XGBoost and LSTM models trained on 200+ behavioral features including session frequency, bet sizing variance, game diversity, and bonus response history. The system produced a calibrated 0–100 risk score with SHAP-based explanations, enabling the CRM team to understand why a player was flagged and craft personalized retention interventions rather than blanket bonuses. Deployed with a real-time inference API (<50ms SLA), Feast feature store, MLflow model registry, and Kafka event streaming, the engine drove a measurable churn rate reduction that translated into a \$15M revenue delta — validated through a rigorous A/B testing framework comparing LTV uplift between treated and holdout cohorts.

\$15M Revenue Impact

2. iGaming Fraud Detection Platform (7 Typologies)

Zensar · iGaming Client · UKGC & MGA Compliant

The iGaming client had no unified fraud detection system — each team operated with siloed rules, fraud was costing 3–5% of Gross Gaming Revenue, and there was no way to detect sophisticated fraud patterns like collusion rings or bot networks in real time. I designed and delivered a multi-model fraud detection platform covering 7 fraud typologies, prioritized by revenue impact and detection complexity. The architecture combined Isolation Forest for anomaly detection, XGBoost with pseudo-label bootstrapping, a Graph Neural Network (GNN) for collusion ring detection using player transaction networks, and an LSTM for sequential bot behavior — all aggregated through a weighted consensus amplifier that produced a calibrated 0–100 risk score. A key product decision was replacing binary fraud flags with tiered actioning (REVIEW / RESTRICT / BLOCK) to reduce false positives on high-value players, protecting revenue while catching fraud. The platform operated on a 3-tier latency architecture (<50ms real-time, 5-minute Flink near-real-time, nightly Airflow batch), with SHAP explanations logged for 100% of BLOCK decisions and SAR filing automated within 24 hours — fully satisfying UKGC and MGA regulatory requirements.

7 Fraud Typologies · Regulatory Compliant · GGR Protection

3. AI Financial Wellness Chatbot — Old Mutual (FAIS-Compliant)

Zensar · Old Mutual · South Africa Financial Services

Old Mutual needed to deliver 24/7 financial guidance at scale, but South Africa's FAIS regulations strictly prohibit AI systems from making financial recommendations — creating a fundamental product tension between user experience and regulatory compliance. I resolved this by defining the "Show Options Rule" as the core product principle: the AI presents 2–3 options with pros and cons, but the user always makes the final decision, keeping the system fully FAIS-compliant. The solution was built on a LangGraph 6-node stateful DAG (Intent Classification → Input Guardrail → RAG Retrieval → Response Generation → Suggestions → Output Guardrail), using Cohere embeddings with pgvector ANN search on Aurora RDS and a dual-layer AWS Bedrock Guardrails stack for PII masking and prompt injection prevention. I chose RAG over fine-tuning because Old Mutual's product documents update frequently — fine-tuning would require expensive retraining on every update, while RAG delivers zero retraining cost with always-current information. A Claude Sonnet 4.5 + Haiku 3 model split routed ~70% of queries to the cheaper model, achieving approximately 10x cost reduction while maintaining response quality — positioning this as the only fully FAIS-compliant AI financial assistant in the market.

FAIS Compliant · 10x Cost Reduction · 24/7 Scale

4. Game Recommendation Engine + Bonus Analysis + Demand Forecasting

Zensar · iGaming Client · Multi-Product AI Portfolio

The iGaming client's player engagement was driven by generic game promotions and untargeted bonus spend, with no data-driven mechanism to match players to games they were likely to enjoy or to forecast demand for capacity and inventory planning. I led the design and delivery of a suite of four interconnected AI products: a hybrid recommendation engine combining collaborative filtering and content-based signals to personalize game discovery; a bonus ROI attribution model using statistical analysis to identify which bonus types drove genuine retention versus short-term exploitation; a demand forecasting system using ARIMA, Prophet, and LSTM models for operational planning; and a shared A/B testing framework to validate each product's impact independently. Each product was scoped with its own KPIs, GTM strategy, and stakeholder sign-off gates, operating on a shared feature store and modular microservices architecture — enabling the team to iterate on each product independently without breaking the others, and collectively improving player engagement, bonus efficiency, and operational predictability across the platform.

Player Engagement · Bonus ROI · Demand Forecasting

5. FDA-Approved AI Medical Scanner

Photon360 · Healthcare · FDA 510(k) Approval

Clinical imaging workflows were burdened by slow, manual interpretation processes that delayed diagnosis and increased the risk of human error — particularly in high-volume settings where radiologists and clinicians were the bottleneck. As co-founder and AI product leader at Photon360, I owned the full product lifecycle of an AI-powered medical scanner: from defining clinical requirements and data pipeline architecture, through model development and clinical validation, to regulatory documentation and FDA approval. The solution used a CNN-based computer vision pipeline for clinical image analysis, validated against clinical ground truth with defined accuracy thresholds agreed upon with clinical partners before development began. The product achieved FDA 510(k) approval — the ultimate product acceptance criterion — making it one of the most rigorous delivery milestones in AI product development, requiring cross-functional alignment between engineering, clinical, and regulatory teams throughout a multi-year development and validation cycle.

FDA 510(k) Approved · Clinical AI · Full Product Lifecycle

6. Autonomous Parking — Volley Automation (Series-A, San Francisco)

Photon360 · Robotics · Series-A Startup

Volley Automation, a Series-A robotics startup in San Francisco, was experiencing significant operational inefficiencies in their autonomous parking system — robot downtime was unplanned, data from sales, engineering, and operations was siloed, and there was no unified data product strategy to support their investor narrative or operational scaling. I defined and delivered a corporate data flow architecture that integrated data streams across sales, engineering, and marketing functions, enabling real-time operational SLA monitoring and predictive maintenance signals for the robot fleet. The solution directly saved 300+ hours of robot downtime by enabling proactive intervention before failures occurred, and the data product strategy was aligned to Volley's Series-A investor narrative — demonstrating operational maturity and scalability to support their next funding round.

300+ Hours Downtime Saved · Series-A Narrative · Robotics AI

7. EdTech Fraud Detection (\$1M Annual Savings)

Photon360 · EdTech Client · \$1M Annual Savings

An EdTech client was losing significant revenue to fraudulent activity — including fake enrollments, payment fraud, and credential abuse — with no systematic detection mechanism in place, relying instead on manual reviews that were slow, inconsistent, and unable to scale with platform growth. I scoped the fraud problem through user and operations research, identifying the highest-impact fraud typologies by financial exposure, and designed an end-to-end detection pipeline combining rule-based heuristics with ML-based anomaly detection and behavioral pattern analysis. The system covered the full pipeline from data ingestion and feature engineering through model inference to an automated alerting system for the operations team. The solution delivered \$1M in annual savings by systematically detecting and blocking fraudulent activity at scale — validated against a pre-defined success metric agreed with the client at project scoping, ensuring the business outcome was measurable and attributable to the AI system.

\$1M Annual Savings · Fraud Prevention · EdTech

8. Whole Slide Imaging (WSI) Digitization Pipeline

Conference · Mayo Clinic · 68% Time Reduction

Mayo Clinic's pathology labs were constrained by a slow, manual biopsy tissue digitization process — each slide required hours of manual handling, metadata transcription, and quality checks, creating a bottleneck that limited lab throughput and introduced transcription errors that could affect patient data integrity. I designed and delivered an end-to-end AI-driven WSI digitization pipeline that automated the entire workflow: from scanner ingestion and image preprocessing through quality assessment and dimensional reduction for gigapixel-scale images, to automated patient metadata mapping from DICOM headers directly to the clinical database — eliminating manual transcription entirely. The pipeline reduced digitization time by 68%, directly increasing pathologist productivity and lab throughput, while the zero-error metadata mapping improved data integrity across Mayo Clinic's clinical database — a critical requirement for downstream AI model training and clinical research.

68% Time Reduction · Zero Transcription Errors · Mayo Clinic

9. Tumor Segmentation (U-Net + Mask R-CNN)

Conference · Mayo Clinic · 99.9% Processing Time Reduction · 66% Cost Reduction

Pathologists at Mayo Clinic were spending hours manually delineating tumor boundaries on gigapixel whole slide images — a process that was not only time-consuming but also subject to inter-observer variability, limiting the consistency and scalability of tumor analysis for research and clinical decision-making. I led the development of an automated tumor segmentation pipeline using a U-Net for pixel-level semantic segmentation combined with Mask R-CNN for instance-level tumor region identification, deployed on a scalable tiling-and-stitching inference architecture that reduced processing time from hours to under one second per WSI — a 99.9% reduction. A critical product decision was to use an ensemble of both models rather than a single approach, prioritizing recall over precision given the clinical cost of a missed tumor region. This AI integration also enabled a 66% reduction in scanner hardware cost compared to competitors, expanding the addressable market from top-tier research centers to mid-tier hospitals and research labs — a 3–4x TAM expansion.

99.9% Faster · 66% Cost Reduction · TAM Expansion

10. TIL Assessment — Breast Cancer Immunotherapy

Conference · Mayo Clinic · Immunotherapy Treatment Targeting

Tumor-Infiltrating Lymphocyte (TIL) assessment is a critical biomarker for determining whether a breast cancer patient is likely to respond to immunotherapy — but manual TIL grading on whole slide images was time-intensive, subjective, and inconsistent across pathologists, creating delays in treatment planning and variability in clinical decisions. I led the development of an automated TIL assessment pipeline that performed cell detection, TIL classification, and density mapping at the whole slide level, producing a standardized TIL score that directly informed oncologists' immunotherapy targeting decisions. The pipeline was integrated into the pathology workflow at Mayo Clinic, reducing time-to-insight for oncologists and replacing subjective manual grading with a reproducible, AI-driven biomarker quantification system — with clinical impact as the primary product KPI, measured by reduction in grading time and concordance with expert pathologist consensus scores.

Immunotherapy Targeting · Reproducible Grading · Clinical Impact

11. PAIP 2020 — MSI Prediction (Ranked 11th Globally)

Onward Health · Pathology AI Challenge · \$2M Seed Funding Contribution

Onward Health needed to establish credibility as a serious AI research company in the digital pathology space to attract clinical partners and seed funding — but as an early-stage startup, it lacked the external validation signals that investors and hospital systems require before committing to a partnership. I framed participation in the PAIP 2020 international pathology AI challenge as a deliberate product strategy: a credibility-building exercise that would generate a published research paper, a global ranking, and a concrete demonstration of technical capability on a real clinical dataset. The solution used a U-Net and EfficientNetB1 ensemble with a custom image stitching strategy for 1 lakh $\times$ 1 lakh pixel colorectal cancer tissue images, achieving a 70% accuracy improvement through the stitching approach on a ~200GB dataset. Ranking 11th globally out of hundreds of international teams, publishing the research paper, and presenting the work to clinical and investor audiences directly contributed to Onward Health's \$2M seed funding round — making this one of the highest-ROI research investments in the company's early history.

11th Global · Published Research · \$2M Seed Funding

12. Skin Lesion Diagnosis (>93% Accuracy)

Onward Health · Dermatology AI · 6-Class Classification

Dermatologists face a high-volume diagnostic challenge with skin lesions — distinguishing between 6 clinically distinct classes requires expert visual interpretation that is time-consuming, subject to variability, and inaccessible in resource-limited settings where specialist dermatologists are scarce. I led the development of a skin lesion diagnosis system using an ensemble of MobileNetV2, DenseNet201, and InceptionV3 architectures combined with a cosine similarity image retrieval network that surfaced the top similar cases from a labeled clinical library — providing pathologists with both a classification output and supporting evidence from analogous cases. The system achieved greater than 93% accuracy across all 6 classes, with accuracy benchmarks co-defined with clinical partners before development to ensure clinical relevance. The product was positioned as a pathologist decision-support tool — not a replacement — which was a deliberate product principle that drove clinical adoption by reducing resistance from the medical community and aligning with regulatory expectations for AI-assisted diagnosis.

>93% Accuracy · 6-Class · Pathologist Decision Support

13. PD-L1 Detection — Lung Adenocarcinoma (Top 3, IIM-B Conference)

Onward Health · Oncology AI · IIM-B Top 3 Project

PD-L1 expression scoring in lung adenocarcinoma is a critical determinant of whether a patient qualifies for immunotherapy — but manual scoring by pathologists is labor-intensive, subjective, and prone to inter-observer variability, leading to inconsistent treatment decisions and delays in therapy initiation. I led the development of a multi-stage PD-L1 detection pipeline: patch extraction from whole slide images, cell detection, PD-L1 positive cell identification, and density scoring to produce a standardized PD-L1 expression score for lung tissue. The system automated a process that previously required hours of expert pathologist time, delivering a reproducible, quantitative output that directly informed oncologists' immunotherapy eligibility decisions. The project was presented to clinical and investor audiences at the IIM-B Conference, where it was recognized as a Top 3 project — contributing to Onward Health's fundraising narrative and establishing the company's credibility in the oncology AI space.

Top 3 IIM-B · Immunotherapy Eligibility · Oncology AI

14. MSI/MSS Biomarker Classification (82% Accuracy)

Onward Health · Colorectal Cancer · Treatment Selection AI

Microsatellite Instability (MSI) and Microsatellite Stability (MSS) status in colorectal carcinoma is a key biomarker that determines treatment selection — MSI-high patients respond well to immunotherapy while MSS patients do not — but traditional molecular testing is expensive, time-consuming, and not universally available, creating a gap in treatment planning for many patients. I led the development of a deep learning-based MSI/MSS classification system using a VGG19 and VGG16 ensemble trained on histopathology patches, enabling MSI status prediction directly from H&E stained tissue slides without requiring molecular testing. The system achieved 82% accuracy on the classification task, with benchmarks co-defined with clinical partners to ensure the output was clinically actionable. By enabling MSI status prediction from standard pathology slides — which are universally available — the product had the potential to democratize access to immunotherapy eligibility testing, particularly in resource-limited settings where molecular testing infrastructure is unavailable.

82% Accuracy · Treatment Selection · Democratized Access

15. OPD Patient Flow Forecasting (30% Inflow Increase)

AlgoLeap · India's Largest Eye Care Chain · 30% Patient Inflow Growth

India's largest eye care hospital chain was operating with unpredictable outpatient department (OPD) patient inflow — leading to chronic over- and under-staffing, long patient wait times, poor resource utilization, and a suboptimal patient experience that was limiting the chain's ability to grow capacity efficiently. I designed and delivered an OPD patient flow forecasting system using a SARIMA and XGBoost ensemble trained on appointment history, seasonal patterns, local events, doctor availability, and holiday calendars — producing multi-horizon forecasts (1-day, 7-day, 30-day) that fed directly into a scheduling recommendation engine and an operations dashboard designed for hospital managers, not data scientists. I directly managed the C-suite client relationship throughout the engagement, co-defining success metrics and presenting monthly ROI reviews. The system drove a 30% increase in patient inflow by enabling proactive scheduling and capacity optimization — and the success of this engagement doubled AlgoLeap's project portfolio as the client expanded the scope to full operations analytics.

30% Patient Inflow Increase · C-Suite Managed · Portfolio Doubled

16. CCTV Analytics — GMR Airport (Number Plate Detection)

AlgoLeap · GMR Airport · \$1M Annual Monetization

GMR Airport's security and operations team was manually reviewing CCTV footage to monitor vehicle movement and detect anomalies — a labor-intensive, reactive process that was unable to scale with the airport's growing traffic volume and provided no real-time operational intelligence for security decision-making. I designed and delivered a real-time CCTV analytics system using YOLO-based object detection for number plate recognition, vehicle counting via centroid tracking algorithms, and an automated alert system for anomalies — all feeding into a live operational intelligence dashboard co-designed with the airport's security and ops teams to ensure the outputs were actionable within their existing workflow. The system was deployed on-premise at GMR Airport to satisfy data sovereignty requirements, using TensorRT-optimized inference for real-time processing of live RTSP camera streams. The solution eliminated redundant manual monitoring workload for the security team and generated \$1M in annual monetization for AlgoLeap — becoming a company-level showcase project and the anchor case for AlgoLeap's airport and infrastructure vertical.

KARVY STOCK BROKING · May 2016 – May 2018 · Data Scientist

17. Stock Momentum Prediction Engine

Karvy Stock Broking · Quantitative Finance · "Wolf of Dalal Street" Award

Karvy's portfolio management team was relying on traditional technical analysis and analyst intuition for stock selection — a process that was inconsistent, difficult to scale across a large portfolio, and unable to systematically exploit momentum patterns that quantitative analysis could detect. I designed and delivered a stock momentum prediction engine using time series analysis (ARIMA), momentum indicators, and statistical hypothesis testing to detect market inefficiencies and generate precise prediction ranges for portfolio performance. The system included a backtesting framework to validate signal quality on historical data before live deployment, and produced actionable prediction ranges — not point estimates — that gave portfolio managers a probabilistic view of expected performance. The engine bridged traditional finance with data science at Karvy, was recognized with the internal "Wolf of Dalal Street" award for its impact on portfolio performance, and laid the quantitative foundation that seeded my data science career trajectory.

"Wolf of Dalal Street" Award · Quantitative Finance · Momentum AI

PROPRIETARY TRADING · May 2014 – May 2016 · Independent Quantitative Trader

18. Cross-Exchange Arbitrage (LME + CME)

Proprietary Trading · London Metal Exchange + Chicago Mercantile Exchange

Price inefficiencies between the London Metal Exchange (LME) and the Chicago Mercantile Exchange (CME) created short-lived arbitrage windows in commodity markets — but exploiting these windows required real-time price feed processing, precise execution timing, and disciplined position sizing to manage risk, all of which were beyond the capability of manual trading approaches. As a self-directed quantitative trader, I designed and built a proprietary rule engine combining rule-based and statistical models to detect arbitrage windows, optimize position sizing, and time execution across both exchanges — effectively acting as my own product owner, defining the strategy, risk parameters, and execution rules from first principles. This experience was foundational: it was my first end-to-end "AI product" — I identified a market problem, designed a data-driven solution, built the system, and managed the outcome with real financial stakes — establishing the quantitative thinking, risk management discipline, and product ownership mindset that has defined my career ever since.

